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LIGHT EMITTING ELEMENT MANUFACTURING METHOD AND LIGHT EMITTING ELEMENT

Abstract

A method of manufacturing a light emitting element includes providing a wafer including a first substrate, a semiconductor structure body, a first electrode, and a second electrode; bonding the second surface side of the semiconductor structure body to a second substrate via a bonding member; subsequently, exposing the first surface of the semiconductor structure body by separating the semiconductor structure body and the first substrate from each other; subsequently, forming a first groove in the semiconductor structure body by removing a portion of the semiconductor structure body, the first groove separating the semiconductor structure body into a plurality of element parts on the second substrate; subsequently, roughening surfaces of the plurality of element parts on the second substrate; and separating the second substrate and the plurality of element parts from each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This is a bypass continuation application of International Application PCT/JP2023/040561, filed on Nov. 10, 2023; the entire contents of which are incorporated herein by reference. This application also claims priority to Japanese Application No. 2022-192110, filed on Nov. 30, 2022, and Japanese Application No. 2023-186715, filed on Oct. 31, 2023. The entire contents of these applications are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a method of manufacturing a light emitting element and a light emitting element.

[0003] A known method of manufacturing a light emitting element includes a step of forming a semiconductor layer on a substrate, a step of bonding the semiconductor layer to another substrate, a step of removing the substrate used to form the semiconductor layer, and a step of roughening a surface of the semiconductor layer exposed by removing the substrate. By roughening the surface of the semiconductor layer, the light extraction efficiency can be increased.

SUMMARY

[0004] One object of certain embodiments of the invention is directed to provide a method of manufacturing a light emitting element and a light emitting element, such that a light emitting element with increased light extraction efficiency can be efficiently manufactured.

[0005] According to an aspect of the invention, a method of manufacturing a light emitting element includes: providing a wafer, the wafer including a first substrate, a semiconductor structure body located on the first substrate, the semiconductor structure body including a first semiconductor layer, a second semiconductor layer located farther from the first substrate than the first semiconductor layer is from the first substrate, and an active layer positioned between the first semiconductor layer and the second semiconductor layer, a first electrode located on a second surface of the semiconductor structure body, the second surface being positioned at a side opposite to a first surface of the semiconductor structure body, the first surface being positioned at a first substrate side, the first electrode being electrically connected with the first semiconductor layer, and a second electrode located on the second surface, the second electrode being electrically connected with the second semiconductor layer; bonding a second surface side of the semiconductor structure body to a second substrate via a bonding member; after the bonding of the semiconductor structure body to the second substrate, exposing the first surface of the semiconductor structure body by separating the semiconductor structure body and the first substrate from each other; after the exposing of the first surface of the semiconductor structure body, forming a first groove in the semiconductor structure body by removing a portion of the semiconductor structure body, the first groove separating the semiconductor structure body into a plurality of element parts on the second substrate; after the forming of the first groove in the semiconductor

structure body, roughening surfaces of the plurality of element parts on the second substrate; and separating the second substrate and the plurality of element parts from each other.

[0006] According to an aspect of the invention, a method of manufacturing a light emitting element includes: providing a wafer, the wafer including a first substrate, a semiconductor structure body located on the first substrate, the semiconductor structure body including a first semiconductor layer, a second semiconductor layer located farther from the first substrate than the first semiconductor layer, and an active layer positioned between the first semiconductor layer and the second semiconductor layer, a first electrode located on a second surface of the semiconductor structure body, the second surface being positioned at a side opposite to a first surface of the semiconductor structure body, the first surface being positioned at a first substrate side, the first electrode being electrically connected with the first semiconductor layer, and a second electrode located on the second surface, the second electrode being electrically connected with the second semiconductor layer; bonding a second surface side of the semiconductor structure body to a second substrate via a bonding member; after the bonding of the semiconductor structure body to the second substrate, exposing the first surface of the semiconductor structure body by separating the semiconductor structure body and the first substrate from each other; after the exposing of the first surface of the semiconductor structure body, roughening the first surface of the semiconductor structure body on the second substrate; after the roughening of the first surface of the semiconductor structure body, forming a first groove in the semiconductor structure body by removing a portion of the semiconductor structure body, the first groove separating the semiconductor structure body into a plurality of element parts on the second substrate; and separating the second substrate and the plurality of element parts from each other.

[0007] According to an aspect of the invention, a light emitting element includes a semiconductor structure body including a first semiconductor layer, a second semiconductor layer, an active layer positioned between the first semiconductor layer and the second semiconductor layer, a first surface, the first surface being a surface of the first semiconductor layer positioned at a side opposite to the active layer, a second surface positioned at a side opposite to the first surface, a lateral surface connecting the first surface and the second surface, the second surface including a first region in which a portion of the first semiconductor layer is exposed from under the second semiconductor layer and the active layer, and a second region being a surface of the second semiconductor layer positioned at a side opposite to the active layer; a first electrode electrically connected with the first semiconductor layer at the first region; and a second electrode electrically connected with the second semiconductor layer at the second region, in which the first surface and the lateral surface are rough surfaces, and an area of the second surface is greater than an area of the first surface.

[0008] According to certain embodiments of the invention, a method of manufacturing a light emitting element and a light emitting element can be provided in which a light emitting element with increased light extraction efficiency can be efficiently manufactured.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 to FIG. 11 are schematic cross-sectional views for describing a step of a method of manufacturing a light emitting element according to a first embodiment;

[0010] FIG. 12 is a schematic plan view for describing a step of the method of manufacturing a light emitting element according to the first embodiment;

[0011] FIG. 13 to FIG. 16 are schematic cross-sectional views for describing a step of the method of manufacturing a light emitting element according to the first embodiment;

[0012] FIG. 17 to FIG. 19 are schematic cross-sectional views for describing a step of a method of

manufacturing a light emitting element according to a second embodiment;

[0013] FIG. **20** to FIG. **22** are schematic cross-sectional views for describing a first modification of the methods for manufacturing the light emitting elements according to the first and second embodiments;

[0014] FIG. **23** and FIG. **24** are schematic cross-sectional views for describing a second modification of the methods for manufacturing the light emitting elements according to the first and second embodiments;

[0015] FIG. **25A** is a schematic plan view of a light emitting element according to an embodiment; and

[0016] FIG. **25B** is a schematic cross-sectional view along line XXVB-XXVB of FIG. **25A**.

DETAILED DESCRIPTION

[0017] Certain embodiments will now be described with reference to the drawings. Unless specifically stated, the dimensions, materials, shapes, relative arrangements, and the like of the components described in the embodiments are not intended to limit the scope of the invention, and are merely illustrative examples. The sizes, positional relationships, and the like of the members shown in the drawings may be exaggerated for clarity of description. Also, in the following description, the same names and reference numerals indicate the same or similar members, and a repeated detailed description is omitted as appropriate. Also, end views that show only cross sections may be used as cross-sectional views. Also, in cross-sectional views, the cross section of the semiconductor structure part is not marked with hatching for easier viewing of the semiconductor structure part.

[0018] In the following description, terms that indicate specific directions or positions (e.g., “up,” “down,” and other terms including such terms) may be used. Such terms, however, are used merely for better understanding of relative directions or positions when referring to the drawings. As long as the relationships are the same, the relative directions or positions according to terms such as “up/above,” “down/below,” etc., used when referring to the drawings may not have the same arrangements in drawings, actual products, and the like outside the disclosure. In the specification, when assuming that there are, for example, two members, the positional relationship expressed as “up (or down)” includes the case where the two members are in contact with each other, and the case in which the two members are not in contact with each other and one of the members is positioned above (or below) the other member. Also, in the specification, unless specifically stated, the expression that a member “covers” an object includes the case where the member is in contact with the object and directly covers the covered object, and the case where the member indirectly covers the covered object without being in contact with the covered object.

First Embodiment

[0019] A method of manufacturing a light emitting element according to a first embodiment will be described with reference to FIGS. **1** to **16**.

<Step of Providing Wafer>

[0020] The method of manufacturing a light emitting element according to the first embodiment includes a step of preparing a wafer. FIG. **7** illustrates a portion of a wafer W. The wafer W includes a first substrate **101**, a semiconductor structure body **10** located on the first substrate **101**, a first electrode **41**, and a second electrode **42**.

[0021] The first substrate **101** can include, for example, an insulating substrate of sapphire or spinel (MgAl.sub.2O.sub.4) having one of a C-plane, an R-plane, or an A-plane as a major surface. Also, a conductive substrate of SiC (including 6H, 4H, and 3C), ZnS, ZnO, GaAs, Si, etc., may be used as the first substrate **101**. According to the present embodiment, a sapphire substrate that has a C-plane as a major surface is used as the first substrate **101**. The semiconductor structure body **10** is formed on the major surface of the first substrate **101**.

[0022] The semiconductor structure body **10** is made of a nitride semiconductor. In the specification, “nitride semiconductor” includes, for example, all compositions of semiconductors

represented by the chemical formula $\text{In}_{\text{sub.x}}\text{Al}_{\text{sub.y}}\text{Ga}_{\text{sub.1-x-y}}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, and $x+y \leq 1$) for which the composition ratios x and y are varied within respective ranges respectively. Also, “nitride semiconductor” further includes nitride semiconductors further containing Group V elements other than N (nitrogen) in the chemical formula above, nitride semiconductors further containing various elements added to control various properties such as the conductivity type, etc. [0023] The semiconductor structure body **10** includes a first semiconductor layer **11**, a second semiconductor layer **13** located at a position farther from the first substrate **101** than the first semiconductor layer **11**, and an active layer **12** positioned between the first semiconductor layer **11** and the second semiconductor layer **13**. The active layer **12** is a light emitting layer that emits light, and has, for example, a MQW (Multiple Quantum well) structure including multiple barrier layers and multiple well layers. For example, the active layer **12** emits light having a peak wavelength of not less than 210 nm and not more than 580 nm. For example, the first semiconductor layer **11** includes a semiconductor layer including an n-type impurity, and the second semiconductor layer **13** includes a semiconductor layer including a p-type impurity.

[0024] The semiconductor structure body **10** includes a first surface **10a** and a second surface **10b**. The first surface **10a** is positioned at the first substrate **101** side. The second surface **10b** is positioned at the side opposite to the first surface **10a**. The second surface **10b** includes a first region **10b1** and a second region **10b2**. In the first region **10b1**, a portion of the first semiconductor layer **11** is exposed from the second semiconductor layer **13** and the active layer **12**. The surface of the second region **10b2** is the upper surface of the second semiconductor layer **13**. The area of the second region **10b2** is greater than the area of the first region **10b1**.

[0025] The active layer **12** is positioned at the side more proximate to the second region **10b2** of the second surface **10b** than the first surface **10a**. The thickness of the semiconductor structure body **10** is, for example, not less than 5 μm and not more than 10 μm . For example, the active layer **12** is positioned within the range of not less than 10 nm and not more than 1,000 nm from the second region **10b2** of the second surface **10b**.

[0026] The first electrode **41** is located on the first region **10b1** of the second surface **10b** and is electrically connected with the first semiconductor layer **11**. The second electrode **42** is located on the second region **10b2** of the second surface **10b** and is electrically connected with the second semiconductor layer **13**.

[0027] The step of providing the wafer **W** may include the steps shown in FIGS. **1** to **6**. The steps of FIGS. **1** to **6** will now be described. In the step of providing the wafer **W**, the wafer **W** shown in FIG. **7** may be provided by purchasing.

[0028] In the step shown in FIG. **1**, the semiconductor structure body **10** is formed on the first substrate **101**. For example, the first semiconductor layer **11**, the active layer **12**, and the second semiconductor layer **13** are formed in this order on the first substrate **101** by MOCVD (Metal Organic Chemical Vapor Deposition). After the semiconductor structure body **10** is formed on the first substrate **101**, a portion of the second semiconductor layer **13** and a portion of the active layer **12** are removed, so that the first region **10b1** and the second region **10b2** are formed in the second surface **10b**. For example, the portion of the second semiconductor layer **13** and the portion of the active layer **12** can be removed by dry etching, wet etching, etc.

[0029] As shown in FIG. **2**, the step of providing the wafer **W** can include forming a first conductive film **21** on the second region **10b2** of the second surface **10b**. The first conductive film **21** is in contact with the upper surface of the second semiconductor layer **13** and has the function of diffusing the current supplied via the second electrode **42** in the in-plane direction of the second semiconductor layer **13**. For example, a light-transmitting conductive film of ITO (Indium Tin Oxide), IZO (Indium Zinc Oxide), ZnO, $\text{In}_{\text{sub.2}}\text{O}_{\text{sub.3}}$, etc., can be used as the first conductive film **21**. For example, the first conductive film **21** can be formed by sputtering and/or vapor deposition.

[0030] As shown in FIG. **6**, the step of providing the wafer **W** can include forming an insulating

film **30**, including a first opening **32a** and a second opening **32b**, at the second surface **10b** of the semiconductor structure body **10**.

[0031] As shown in FIG. **2**, the step of forming the insulating film **30** includes forming a first film **31** at the second surface **10b** of the semiconductor structure body **10**. The first film **31** covers the second surface **10b** and the first conductive film **21**.

[0032] The first film **31** can be a film that is reflective to the light emitted by the active layer **12**. The reflectance of the first film **31** for the light emitted by the active layer **12** is not less than 60%, and favorably not less than 70%. The first film **31** includes, for example, a dielectric multilayer film. The dielectric multilayer film includes, for example, a SiO.sub.2 layer and a Nb.sub.2O.sub.5 layer that are alternately stacked. It is favorable to form the first film **31**, for example, by forming a relatively thick SiO.sub.2 layer having a thickness of not less than 100 nm and not more than 500 nm, and subsequently forming not less than two and not more than six pairs of a Nb.sub.2O.sub.5 layer having a thickness of not less than 10 nm and not more than 100 nm and a SiO.sub.2 layer having a thickness of not less than 10 nm and not more than 100 nm as a dielectric multilayer film on the relatively thick SiO.sub.2 layer. Setting the film thickness of each layer and the number of stacks of the layers of the first film **31** within these ranges allows for having good light reflectivity. For example, as the first film **31**, a SiO.sub.2 layer having a thickness of 300 nm can be formed, and subsequently three pairs of a Nb.sub.2O.sub.5 layer having a thickness of 52 nm and a SiO.sub.2 layer having a thickness of 83 nm can be formed on the SiO.sub.2 layer of the thickness of 300 nm. A material such as titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), etc., can be used as the material of the first film **31**. For example, the first film **31** can be formed by CVD (Chemical Vapor Deposition) and/or sputtering.

[0033] As shown in FIG. **3**, the step of forming the insulating film **30** includes forming a fourth opening **31a** and a fifth opening **31b** in the first film **31**. The first region **10b1** of the second surface **10b**, i.e., the first semiconductor layer **11**, is exposed in the fourth opening **31a** of the first film **31**. The first conductive film **21** is exposed in the fifth opening **31b** of the first film **31**. For example, the fourth opening **31a** and the fifth opening **31b** can be formed by removing portions of the first film **31** by dry etching and/or wet etching.

[0034] As shown in FIG. **4**, the step of preparing the wafer **W** can include a step of forming a second conductive film **22**, a third conductive film **23**, and a reflective film **24**. For example, the second conductive film **22**, the third conductive film **23**, and the reflective film **24** can be formed by sputtering and/or vapor deposition.

[0035] The second conductive film **22** is located on the first semiconductor layer **11** in the fourth opening **31a**, and is electrically connected with the first semiconductor layer **11**. The second conductive film **22** allows for reducing the contact resistance between the first electrode **41** and the first semiconductor layer **11**.

[0036] The third conductive film **23** is located on the first conductive film **21** in the fifth opening **31b**, and is electrically connected with the first conductive film **21**. The third conductive film **23** allows for reducing the contact resistance between the second electrode **42** and the first conductive film **21**.

[0037] The second conductive film **22** and the third conductive film **23** can be, for example, a single-layer metal layer including Ti, Rh, Au, Pt, Al, Ag, or Ru, or a stacked structure including at least two among these metal layers. The second conductive film **22** and the third conductive film **23** can be simultaneously formed by using the same material.

[0038] The reflective film **24** is formed on the upper surface of the first film **31**. The reflective film **24** is reflective to the light emitted by the active layer **12**. For example, a metal can be used as the material of the reflective film **24**. The reflective film **24** includes, for example, an Al film, a Ti film, or a stacked structure of an Al film and a Ti film.

[0039] As shown in FIG. **5**, the step of forming the insulating film **30** further includes a step of

forming a second film **32** on the first film **31**. The second film **32** covers the semiconductor structure body **10**, the first conductive film **21**, the second conductive film **22**, the third conductive film **23**, and the reflective film **24**. For example, SiO.sub.2, SiN, SiON, etc., can be used as the second film **32**. For example, the second film **32** can be formed by CVD and/or sputtering.

[0040] After the second film **32** is formed, the first opening **32a** and the second opening **32b** are formed in the second film **32** as shown in FIG. **6**. In the first opening **32a**, the second conductive film **22** is exposed from under the insulating film **30**. In the second opening **32b**, the third conductive film **23** is exposed from under the insulating film **30**.

[0041] As shown in FIG. **7**, the first electrode **41** is located in the first opening **32a**, and the first electrode **41** is in contact with the second conductive film **22**. The first electrode **41** is electrically connected with the first semiconductor layer **11** via the second conductive film **22**. The second electrode **42** is located in the second opening **32b**, and the second electrode **42** is in contact with the third conductive film **23**. The second electrode **42** is electrically connected with the second semiconductor layer **13** via the third and first conductive films **23** and **21**. The first electrode **41** and the second electrode **42** include, for example, a Ti layer, a Rh layer, a Au layer, or a stacked structure of any two of the Ti layer, the Rh layer, or the Au layer. The first electrode **41** and the second electrode **42** can be simultaneously formed using the same material. The first electrode **41** and the second electrode **42** can be formed by sputtering and/or vapor deposition.

[0042] After the step of preparing the wafer **W**, as shown in FIG. **8**, the method of manufacturing a light emitting element according to the first embodiment includes a step of bonding the second surface **10b** side of the semiconductor structure body **10** to a second substrate **102** via a bonding member **50**. In the illustration of FIG. **8**, the vertical positions of the wafer **W** are inverted from the drawings up to FIG. **7**.

[0043] The bonding member **50** is located between the insulating film **30** and the second substrate **102**. Also, the bonding member **50** covers the first electrode **41** and the second electrode **42**. The bonding member **50** is, for example, a resin member that mainly includes an epoxy resin, an acrylic resin, or a polyimide resin. For example, similarly to the first substrate **101**, a substrate of sapphire, spinel, SiC, ZnS, ZnO, GaAs, Si, etc., can be used as the second substrate **102**.

[0044] The method of manufacturing a light emitting element according to the first embodiment includes, after the step of bonding the semiconductor structure body **10** to the second substrate **102**, a step of exposing the first surface **10a** of the semiconductor structure body **10** as shown in FIG. **9** by separating the semiconductor structure body **10** and the first substrate **101** from each other.

[0045] For example, the first substrate **101** is removed by a method such as LLO (Laser Lift Off), grinding, polishing, etching, etc.

[0046] The method of manufacturing a light emitting element according to the first embodiment includes, after the step of exposing the first surface **10a** of the semiconductor structure body **10**, a step of forming a first groove **71** shown in FIG. **11** in the semiconductor structure body **10** by removing a portion of the semiconductor structure body **10**.

[0047] As shown in FIGS. **11** and **12**, the first groove **71** separates the semiconductor structure body **10** into multiple element parts **100** on the second substrate **102**. The cross section of FIG. **11** illustrates a XI-XI cross section of FIG. **12**.

[0048] In the step of forming the first groove **71**, for example, the first groove **71** is formed by removing a portion of the semiconductor structure body **10** by dry etching using a mask **61** arranged on the first surface **10a** as shown in FIG. **10**. The mask **61** is, for example, a resist mask. Examples of dry etching include RIE (Reactive Ion Etching). For example, a gas that includes chlorine such as Cl.sub.2, SiCl.sub.4, etc., can be used in the RIE. Alternatively, the first groove **71** may be formed by removing a portion of the semiconductor structure body **10** by wet etching. Compared with wet etching, the etching amount of dry etching can be easily controlled, which can facilitate stabilizing the shape of the element part **100** after etching.

[0049] The element part **100** includes the first surface **10a**, the second surface **10b**, and lateral

surfaces **10c** that connect the first surface **10a** and the second surface **10b**. The light that is emitted by the active layer **12** is extracted outside the semiconductor structure body **10** mainly from the first surface **10a** and the lateral surfaces **10c**.

[0050] In the step of forming the first groove **71** in the semiconductor structure body **10**, the first groove **71** is formed to expose a portion **30a** of a surface of the insulating film **30** from under the semiconductor structure body **10**. The first groove **71** is defined by lateral surfaces **10c** of element parts **100** and the portion **30a** of the surface of the insulating film **30**. As shown in FIG. **11**, the lateral surface **10c** is inclined with respect to the first and second surfaces **10a** and **10b**. The angle between the lateral surface **10c** and the second surface **10b** (the interior angle formed inside the semiconductor structure body **10** by the lateral surface **10c** and the second surface **10b**) is an acute angle, and the angle between the lateral surface **10c** and the first surface **10a** (the interior angle formed inside the semiconductor structure body **10** by the lateral surface **10c** and the first surface **10a**) is an obtuse angle. The active layer **12** is positioned at the side more proximate to the second surface **10b** than the first surface **10a**; therefore, when the angle between the lateral surface **10c** and the second surface **10b** is an acute angle and the angle between the lateral surface **10c** and the first surface **10a** is an obtuse angle, the active layer **12** of the element part **100** can have a large area. By forming the first groove **71** in the semiconductor structure body **10** by etching, the angle between the lateral surface **10c** and the second surface **10b** can be an acute angle, and the angle between the lateral surface **10c** and the first surface **10a** can be an obtuse angle. As shown in FIG. **11**, the cross-sectional shape of the element part **100** is substantially trapezoidal.

[0051] In the step of forming the first groove **71**, which separates the semiconductor structure body **10** into the multiple element parts **100**, by etching, the time during which a portion of the semiconductor structure body **10** at the first surface **10a** side is exposed to the etching gas or the etchant is longer than the time during which a portion of the semiconductor structure body **10** at the second surface **10b** side is exposed to the etching gas or the etchant. As a result, there is a tendency for the width of the first surface **10a** side of the first groove **71** to be greater than the width of the second surface **10b** side. Accordingly, the cross-sectional shape of the semiconductor structure body **10** adjacent to the first groove **71** is substantially trapezoidal.

[0052] As a comparative example, when the semiconductor structure body **10** is separated into multiple element parts by forming a groove in the semiconductor structure body **10** positioned on the first substrate **101** before removing the first substrate **101**, the etching of the semiconductor structure body **10** proceeds from the second surface **10b** side. In such a case, the cross-sectional shape of the semiconductor structure body **10** is a substantially trapezoidal shape in which the area of the first surface **10a** is greater than the area of the second surface **10b**. The active layer **12** is positioned at the side that is more proximate to the second surface **10b** than the first surface **10a**, so that the area of the active layer **12** tends to be reduced in the case of a substantially trapezoidal shape in which the area of the first surface **10a** is greater than the area of the second surface **10b**.

[0053] According to the present embodiment, within the active layer **12** of the semiconductor structure body **10** that is positioned on the first substrate **101** before removing the first substrate **101**, removal is performed only in the first region **10b1**, on which the first electrode **41** is to be arranged. In a state in which the semiconductor structure body **10** is positioned on the first substrate **101**, a groove that separates the semiconductor structure body **10** into multiple element parts has not been formed. After the semiconductor structure body **10** is bonded to the second substrate **102**, the first substrate **101** is removed, and the first groove **71** is formed by causing the etching to proceed from the first surface **10a** side. Thus, the cross-sectional shape of the semiconductor structure body **10** can be a substantially trapezoidal shape in which the area of the second surface **10b** at the lower side is greater than the area of the first surface **10a** at the upper side. Accordingly, compared to a case in which the semiconductor structure body **10** is separated into the element parts **100** on the first substrate **101**, the area of the active layer **12** can be increased, and the light extraction efficiency can be increased.

[0054] The method of manufacturing a light emitting element according to the first embodiment includes, after the step of forming the first groove **71** in the semiconductor structure body **10**, the step of roughening the surfaces of the multiple element parts **100** on the second substrate **102** as shown in FIG. **13**.

[0055] The first surface **10a** and the lateral surfaces **10c** of the element part **100** are roughened. The roughened first surface **10a** and lateral surfaces **10c** include multiple protrusions. By roughening the first surface **10a** and the lateral surfaces **10c**, the light extraction efficiency from the first surface **10a** and the lateral surface **10c** can be increased. For example, the first surface **10a** and the lateral surface **10c** are roughened by dry etching using a gas including chlorine, or wet etching using an alkaline solution such as TMAH (Tetramethylammonium hydroxide), etc. Thus, the first surface **10a** and the lateral surface **10c** can be roughened in a single step, so that a light emitting element with increased light extraction efficiency can be efficiently manufactured.

[0056] According to the first embodiment, the first groove **71** is formed in the semiconductor structure body **10** by removing a portion of the semiconductor structure body **10** in a state in which the semiconductor structure body **10** shown in FIG. **11** above is not roughened. Therefore, compared to when separating into the element parts **100** by removing a portion of the semiconductor structure body **10** after roughening the first surface **10a** of the semiconductor structure body **10**, the first surface **10a** of the semiconductor structure body **10** in a highly flat state can be etched and thus the first groove **71** can be formed, which can facilitate stabilizing the shape of the element part **100**.

[0057] Also, by performing the roughening step after separating into the element parts **100**, the steps can be simplified, and not only the first surface **10a** but also the lateral surface **10c** can be roughened. As a result, the light extraction efficiency from the semiconductor structure body **10** can be increased. Also, when the semiconductor structure body **10** is grown on the C-plane of a sapphire substrate used as the first substrate **101**, due to difference between the crystal orientation of the first surface **10a** and that of the lateral surface **10c**, the surface roughness of the first surface **10a** can be greater than the surface roughness of the lateral surface **10c** by performing a roughening step after separating into the element parts **100** as shown in FIG. **13**. With the surface roughness of the first surface **10a** greater than the surface roughness of the lateral surface **10c**, the light is extracted more easily from the first surface **10a** than the lateral surface **10c**, so that light distribution characteristics that have high directivity in the direction directly above the first surface **10a** can be obtained. For example, the surface roughness of the first surface **10a** and the surface roughness of the lateral surface **10c** can be expressed by the maximum height R_z . Also, for example, the surface roughness of the first surface **10a** and the surface roughness of the lateral surface **10c** can be expressed by the arithmetic average roughness R_a . For example, the maximum height R_z of the first surface **10a** is not less than $0.5\ \mu\text{m}$ and not more than $3.0\ \mu\text{m}$, and the maximum height R_z of the lateral surface **10c** is not less than $10\ \text{nm}$ and not more than $400\ \text{nm}$. Also, for example, the arithmetic average roughness R_a of the first surface **10a** is not less than $100\ \text{nm}$ and not more than $300\ \text{nm}$, and the arithmetic average roughness R_a of the lateral surface **10c** is not less than $1\ \text{nm}$ and not more than $100\ \text{nm}$. For example, the surface roughness of the first surface **10a** and the surface roughness of the lateral surface **10c** can be measured with a laser microscope, an atomic force microscope, etc.

[0058] the method of manufacturing a light emitting element according to the first embodiment can include, after the step of roughening the first surface **10a** and the lateral surface **10c**, a step of forming a protective film **80** covering the first surface **10a** and the lateral surface **10c** as shown in FIG. **14**. Also, the protective film **80** covers the portion **30a** of the surface of the insulating film **30**. For example, the protective film **80** can be formed by CVD and/or sputtering.

[0059] A shape that follows the rough surface shapes of the first surface **10a** and the lateral surface **10c** is formed in the surface of the protective film **80**. As a result, the extraction efficiency of the light extracted from the first surface **10a** and the lateral surface **10c** via the protective film **80** can

be increased. The protective film **80** is transmissive to the light emitted by the active layer **12**. The transmittance of the protective film **80** for the light emitted by the active layer **12** is not less than 60%, and favorably not less than 70%. For example, SiO₂, SiN, and SiON can be used as the protective film **80**.

[0060] The method of manufacturing a light emitting element according to the first embodiment can further include, after the step of forming the first groove **71**, a step of forming a second groove **72** in the insulating film **30** and the bonding member **50** as shown in FIG. **15** by removing a corresponding portion of the insulating film **30** and a corresponding portion of the bonding member **50** below the first groove **71**.

[0061] According to the present embodiment, after the first groove **71** is formed, the first surface **10a** and the lateral surface **10c** are roughened; the protective film **80** is then formed; subsequently, the second groove **72** is formed. For example, the second groove **72** is formed by removing the portion of the insulating film **30** and the portion of the bonding member **50** below the first groove **71** by RIE using a mask. For example, a gas that includes fluorine such as CF₄, CHF₃, or the like is used when removing the insulating film **30**, and a gas that includes oxygen such as O₂ or the like is used when removing the bonding member **50**.

[0062] The second groove **72** reaches a surface of the second substrate **102**. The insulating film **30** is separated by the second groove **72** into multiple portions corresponding to the multiple element parts **100**. As shown in FIG. **15**, in the element part **100**, the end portion of the insulating film **30** and the end portion of the second surface **10b** are formed to be aligned with each other.

Accordingly, when the first film **31** is reflective to the light emitted by the active layer **12**, the light that is emitted by the active layer **12** can be efficiently reflected. The bonding member **50** is also separated by the second groove **72** into multiple portions corresponding to the multiple element parts **100**. The element parts **100** each are supported on the second substrate **102** via the bonding member **50** with the second surface **10b** facing the second substrate **102**.

[0063] The method of manufacturing a light emitting element according to the first embodiment includes a step of separating the second substrate **102** and the multiple element parts **100** from each other.

[0064] For example, the element part **100** and the second substrate **102** can be separated from each other by removing the bonding member **50** by irradiating laser light from the second substrate **102** side. As shown in FIG. **16**, the first surface **10a** of the element part **100** that is separated from the second substrate **102** is bonded to, for example, an adhesive support member **103** via the protective film **80**. The element part **100** may be separated from the second substrate **102** after being bonded to the support member **103**. After the element part **100** is separated from the second substrate **102**, the first electrode **41** and the second electrode **42** are exposed by removing the remaining portion of the bonding member **50** at the insulating film **30** side by, for example, RIE. Thus, a light emitting element **1** is obtained. The exposed first and second electrodes **41** and **42** function as external connection terminals bonded to a mounting substrate. The light emitting element **1** is, for example, a light emitting diode.

Second Embodiment

[0065] A method of manufacturing a light emitting element according to a second embodiment will be described with reference to FIGS. **17** to **19**.

[0066] Similarly to the first embodiment, the method of manufacturing a light emitting element according to the second embodiment includes a step of providing the wafer **W**, a step of bonding the second surface **10b** side of the semiconductor structure body **10** to the second substrate **102** via the bonding member **50**, and a step of exposing the first surface **10a** of the semiconductor structure body **10** by separating the semiconductor structure body **10** and the first substrate **101** from each other.

[0067] The method of manufacturing a light emitting element according to the second embodiment includes, after the step of exposing the first surface **10a** of the semiconductor structure body **10**, a

step of roughening the first surface **10a** of the semiconductor structure body **10** on the second substrate **102** as shown in FIG. **17**.

[0068] Similarly to the first embodiment, for example, the first surface **10a** is roughened by dry etching using a gas including chlorine, or by wet etching using an alkaline solution such as TMAH, etc.

[0069] The method of manufacturing a light emitting element according to the second embodiment includes, after the step of roughening the first surface **10a** of the semiconductor structure body **10**, a step of forming, in the semiconductor structure body **10**, the first groove **71** that separates the semiconductor structure body **10** into the multiple element parts **100** on the second substrate **102** by removing a portion of the semiconductor structure body **10**.

[0070] In the step of forming the first groove **71**, for example, the first groove **71** is formed by removing a portion of the semiconductor structure body **10** by dry etching using the mask **61** that is shown in FIG. **18** and is located on the first surface **10a**. Examples of dry etching include RIE. For example, a gas that includes chlorine such as Cl.sub.2, SiCl.sub.4, etc., can be used in the RIE. Because the etching controllability of dry etching is good, the shape of the element part **100** after etching is easily stabilized.

[0071] The dry etching of the semiconductor structure body **10** proceeds from the roughened first surface **10a** exposed from under the mask **61**. Therefore, as shown in FIG. **19**, the surface roughness of the surface subjected to dry etching is greater than when the first surface **10a** is pre-roughened. Accordingly, the lateral surface **10c** of the element part **100** exposed in the step of forming the first groove **71** is roughened as shown in FIG. **13**. As a result, the formation of the first groove **71** and the roughening of the lateral surface **10c** can be performed in a single step, so that a light emitting element with increased light extraction efficiency can be efficiently manufactured. Also, when forming the first groove **71** by performing dry etching of the pre-roughened first surface **10a**, the state of the surface of the lateral surface **10c** is easily controlled. For example, compared to a case in which the first surface **10a** and the lateral surface **10c** are roughened after the first groove **71** is formed, the difference between the surface roughness of the first surface **10a** and the surface roughness of the lateral surface **10c** is easily reduced.

[0072] Also, the first groove **71** may be formed by removing a portion of the semiconductor structure body **10** by wet etching. Compared to dry etching, when performing wet etching, the surface roughness of the lateral surface **10c** is easily increased, and the light extraction efficiency from the lateral surface **10c** is easily increased.

[0073] According to the method of manufacturing a light emitting element according to the second embodiment, the light emitting element **1** is obtained by performing the step of forming the first groove **71** described above and then steps that is the same as those of the first embodiment shown in FIGS. **14** to **16** above.

[0074] FIGS. **20** to **22** are schematic cross-sectional views for describing a first modification of the methods for manufacturing the light emitting elements according to the first and second embodiments.

[0075] According to the first modification, the step of providing the wafer W further includes forming the insulating film **30** that includes the first opening **32a**, the second opening **32b**, and a third opening **32c** in the second surface **10b** of the semiconductor structure body **10** as shown in FIG. **20**. The third opening **32c** extends through the second and first films **32** and **31** and reaches the second region **10b2** of the second surface **10b** (the upper surface of the second semiconductor layer **13**). For example, the first opening **32a**, the second opening **32b**, and the third opening **32c** are formed simultaneously by RIE.

[0076] In the step of bonding the semiconductor structure body **10** to the second substrate **102** via the bonding member **50**, a portion **50a** of the bonding member **50** is located in the third opening **32c** as shown in FIG. **21**. For example, the portion **50a** of the bonding member **50** is in contact with the upper surface of the second semiconductor layer **13** in the third opening **32c**.

[0077] Then, in the step of forming the first groove **71**, the first groove **71** is formed above the third opening **32c** as shown in FIG. **22**. According to the first embodiment, the first surface **10a** and the lateral surface **10c** of the element part **100** are roughened after the first groove **71** is formed. According to the second embodiment, as described above, the first groove **71** is formed after the first surface **10a** is roughened, and thus the lateral surface **10c** can be roughened in the step of forming the first groove **71**.

[0078] In the step of forming the insulating film **30** according to the first modification, the insulating film **30** is divided by the third opening **32c**. As described above, the third opening **32c** can be formed simultaneously with the first opening **32a** in which the first electrode **41** will be located and the second opening **32b** in which the second electrode **42** will be disposed. The portion **50a** of the bonding member **50** located in the third opening **32c** can be removed when removing the bonding member **50** in the step of separating the element part **100** and the second substrate **102** from each other. As a result, as shown in FIG. **16**, the multiple light emitting elements **1** that are separated from each other are obtained. Accordingly, in the first modification, the necessity of the step of forming the second groove **72** after forming the first groove **71** is eliminated, so that the light emitting element can be manufactured efficiently.

[0079] FIGS. **23** and **24** are schematic cross-sectional views for describing a second modification of the methods for manufacturing the light emitting elements according to the first and second embodiments.

[0080] In the second modification, before the step of forming the protective film **80** covering the first surface **10a** and the lateral surface **10c**, the second groove **72** is formed by removing a portion of the insulating film **30** and a portion of the bonding member **50** below the first groove **71** as shown in FIG. **23**. In this step of forming the second groove **72**, the portion of the insulating film **30** and the portion of the bonding member **50** can be removed using the semiconductor structure body **10** as a mask. Accordingly, it is unnecessary to form another mask when forming the second groove **72**, so that the light emitting element can be manufactured efficiently.

[0081] The second modification further includes, after the step of forming the second groove **72**, a step of forming the protective film **80** that covers the semiconductor structure body **10** and the lateral surfaces of the insulating film **30** defining the second groove **72** as shown in FIG. **24**. The protective film **80** covers the first surface **10a** and the lateral surfaces **10c** of the element part **100**. The protective film **80** also covers the lateral surfaces of the bonding member **50** defining the second groove **72**. Also, the protective film **80** is located at the surface of the second substrate **102** positioned at the bottom of the second groove **72**.

[0082] Subsequently, multiple light emitting elements that are separated from each other are obtained by removing the bonding member **50** in the step of separating the second substrate **102** and the element parts **100** from each other. In the light emitting elements that are obtained according to the second modification, the lateral surfaces of the insulating film **30** is covered with the protective film **80**.

Light Emitting Element

[0083] The light emitting element **1** according to an embodiment will now be described with reference to FIGS. **25A** and **25B**.

[0084] As described above, the light emitting element **1** includes the semiconductor structure body **10**, the first electrode **41**, and the second electrode **42**.

[0085] The semiconductor structure body **10** includes the first semiconductor layer **11**, the second semiconductor layer **13**, and the active layer **12** positioned between the first semiconductor layer **11** and the second semiconductor layer **13**. The semiconductor structure body **10** also includes the first surface **10a**, the second surface **10b** positioned at a side opposite to the first surface **10a**, and the lateral surfaces **10c** connecting the first surface **10a** and the second surface **10b**. The first surface **10a** is a surface of the first semiconductor layer **11** positioned at the side opposite to the active layer **12**. The second surface **10b** includes the first region **10b1** in which a portion of the first

semiconductor layer **11** is exposed from under the second semiconductor layer **13** and the active layer **12**, and the second region **10b2** that is a surface of the second semiconductor layer **13** positioned at a side opposite to the active layer **12**. The first surface **10a** and the lateral surfaces **10c** are rough surfaces. As shown in FIG. 25A, when viewed in plan, the shape of the semiconductor structure body **10** is quadrilateral and includes four lateral surfaces **10c**. All of the four lateral surfaces **10c** are rough surfaces. When the first surface **10a** and the lateral surfaces **10c** are rough surfaces, the light extraction efficiency from the first surface **10a** and the lateral surfaces **10c** can be increased.

[0086] As shown in FIG. 25B, the cross-sectional shape of the semiconductor structure body **10** is substantially trapezoidal, and the area of the second surface **10b** at the lower side is greater than the area of the first surface **10a** at the upper side. As a result, the area of the active layer **12** positioned more proximate to the second surface **10b** than the first surface **10a** can be increased, and thus the light extraction efficiency can be increased.

[0087] The surface roughness of the first surface **10a** is greater than the surface roughness of the lateral surface **10c**. As a result, light is extracted more easily from the first surface **10a** than the lateral surface **10c**, which can result in light distribution characteristics in which directivity is higher in the direction toward a location directly above the first surface **10a**. As described above, for example, the surface roughness of the first surface **10a** and the surface roughness of the lateral surface **10c** can be expressed by the maximum height R_z . Also, for example, the surface roughness of the first surface **10a** and the surface roughness of the lateral surface **10c** can be expressed by the arithmetic average roughness R_a . For example, the maximum height R_z of the first surface **10a** is not less than 0.5 μm and not more than 3.0 μm , and the maximum height R_z of the lateral surface **10c** is not less than 10 nm and not more than 400 nm. Also, for example, the arithmetic average roughness R_a of the first surface **10a** is not less than 100 nm and not more than 300 nm, and the arithmetic average roughness R_a of the lateral surface **10c** is not less than 1 nm and not more than 100 nm.

[0088] The first electrode **41** is electrically connected with the first semiconductor layer **11** at the first region **10b1**. The second electrode **42** is electrically connected with the second semiconductor layer **13** at the second region **10b2**.

[0089] The light emitting element **1** can further include the first conductive film **21**, the second conductive film **22**, the third conductive film **23**, the reflective film **24**, the insulating film **30**, and the protective film **80** described above.

[0090] Hereinabove, embodiments of the invention are described with reference to specific examples. However, the invention is not limited to these specific examples. All configurations practicable by an appropriate design modification by one skilled in the art based on the embodiments of the invention described above also are within the scope of the invention to the extent that the purport of the invention is included. Furthermore, various modifications and alterations within the spirit of the invention will be readily apparent to those skilled in the art, and all such modifications and alterations also are within the scope of the invention.

Claims

1. A method of manufacturing a light emitting element, comprising: providing a wafer, the wafer comprising: a first substrate, a semiconductor structure body located on the first substrate, the semiconductor structure body having a first surface at a first substrate side, and a second surface at a side opposite the first surface, the semiconductor structure body comprising: a first semiconductor layer, a second semiconductor layer located farther from the first substrate than the first semiconductor layer is from the first substrate, and an active layer positioned between the first semiconductor layer and the second semiconductor layer, a first electrode located on the second surface of the semiconductor structure body, the first electrode being electrically connected with

the first semiconductor layer, and a second electrode located on the second surface of the semiconductor structure body, the second electrode being electrically connected with the second semiconductor layer; bonding a second surface side of the semiconductor structure body to a second substrate via a bonding member; after the bonding of the semiconductor structure body to the second substrate, exposing the first surface of the semiconductor structure body by separating the semiconductor structure body and the first substrate from each other; after the exposing of the first surface of the semiconductor structure body, forming a first groove in the semiconductor structure body by removing a portion of the semiconductor structure body, the first groove separating the semiconductor structure body into a plurality of element parts on the second substrate; after the forming of the first groove in the semiconductor structure body, roughening surfaces of the plurality of element parts on the second substrate; and separating the second substrate and the plurality of element parts from each other.

2. A method of manufacturing a light emitting element, comprising: providing a wafer, the wafer comprising: a first substrate, a semiconductor structure body located on the first substrate, the semiconductor structure body having a first surface at a first substrate side, and a second surface at a side opposite the first surface, the semiconductor structure body comprising: a first semiconductor layer, a second semiconductor layer located farther from the first substrate than the first semiconductor layer is from the first substrate, and an active layer positioned between the first semiconductor layer and the second semiconductor layer, a first electrode located on the second surface of the semiconductor structure body, the first electrode being electrically connected with the first semiconductor layer, and a second electrode located on the second surface of the semiconductor structure body, the second electrode being electrically connected with the second semiconductor layer; bonding a second surface side of the semiconductor structure body to a second substrate via a bonding member; after the bonding of the semiconductor structure body to the second substrate, exposing the first surface of the semiconductor structure body by separating the semiconductor structure body and the first substrate from each other; after the exposing of the first surface of the semiconductor structure body, roughening the first surface of the semiconductor structure body while the semiconductor structure body is on the second substrate; after the roughening of the first surface of the semiconductor structure body, forming a first groove in the semiconductor structure body by removing a portion of the semiconductor structure body, the first groove separating the semiconductor structure body into a plurality of element parts on the second substrate; and separating the second substrate and the plurality of element parts from each other.

3. The method of manufacturing a light emitting element according to claim 2, wherein: in the forming of the first groove in the semiconductor structure body, the first groove is formed in the semiconductor structure body by removing the portion of the semiconductor structure body from the roughened first surface side by dry etching.

4. The method of manufacturing a light emitting element according to claim 1, wherein: the providing of the wafer further comprises forming an insulating film at the second surface of the semiconductor structure body, and in the forming of the first groove in the semiconductor structure body, the first groove is formed to expose the insulating film from under the semiconductor structure body.

5. The method of manufacturing a light emitting element according to claim 2, wherein: the providing of the wafer further comprises forming an insulating film at the second surface of the semiconductor structure body, and in the forming of the first groove in the semiconductor structure body, the first groove is formed to expose the insulating film from under the semiconductor structure body.

6. The method of manufacturing a light emitting element according to claim 3, wherein: the providing of the wafer further comprises forming an insulating film at the second surface of the semiconductor structure body, and in the forming of the first groove in the semiconductor structure body, the first groove is formed to expose the insulating film from under the semiconductor

structure body.

7. The method of manufacturing a light emitting element according to claim 4, further comprising: after the forming of the first groove, forming a second groove in the insulating film and in the bonding member by removing a portion of the insulating film and a portion of the bonding member that are located below the first groove.
8. The method of manufacturing a light emitting element according to claim 5, further comprising: after the forming of the first groove, forming a second groove in the insulating film and in the bonding member by removing a portion of the insulating film and a portion of the bonding member that are located below the first groove.
9. The method of manufacturing a light emitting element according to claim 6, further comprising: after the forming of the first groove, forming a second groove in the insulating film and in the bonding member by removing a portion of the insulating film and a portion of the bonding member that are located below the first groove.
10. The method of manufacturing a light emitting element according to claim 1, wherein: the providing of the wafer further comprises forming an insulating film on the second surface of the semiconductor structure body, the insulating film includes a first opening, a second opening, and a third opening, in the providing of the wafer, the first electrode is disposed in the first opening, and the second electrode is disposed in the second opening, and in the forming of the first groove, the first groove is formed above the third opening.
11. The method of manufacturing a light emitting element according to claim 2, wherein: the providing of the wafer further comprises forming an insulating film on the second surface of the semiconductor structure body, the insulating film includes a first opening, a second opening, and a third opening, in the providing of the wafer, the first electrode is disposed in the first opening, and the second electrode is disposed in the second opening, and in the forming of the first groove, the first groove is formed above the third opening.
12. The method of manufacturing a light emitting element according to claim 3, wherein: the providing of the wafer further comprises forming an insulating film on the second surface of the semiconductor structure body, the insulating film includes a first opening, a second opening, and a third opening, in the providing of the wafer, the first electrode is disposed in the first opening, and the second electrode is disposed in the second opening, and in the forming of the first groove, the first groove is formed above the third opening.
13. The method of manufacturing a light emitting element according to claim 7, wherein: in the forming of the second groove, the portion of the insulating film and the portion of the bonding member are removed by using the semiconductor structure body as a mask.
14. The method of manufacturing a light emitting element according to claim 8, wherein: in the forming of the second groove, the portion of the insulating film and the portion of the bonding member are removed by using the semiconductor structure body as a mask.
15. The method of manufacturing a light emitting element according to claim 9, wherein: in the forming of the second groove, the portion of the insulating film and the portion of the bonding member are removed by using the semiconductor structure body as a mask.
16. The method of manufacturing a light emitting element according to claim 13, further comprising: forming a protective film after the forming of the second groove, the protective film covering the semiconductor structure body, and a lateral surface of the insulating film defining the second groove.
17. The method of manufacturing a light emitting element according to claim 14, further comprising: forming a protective film after the forming of the second groove, the protective film covering the semiconductor structure body, and a lateral surface of the insulating film defining the second groove.
18. The method of manufacturing a light emitting element according to claim 15, further comprising: forming a protective film after the forming of the second groove, the protective film

covering the semiconductor structure body, and a lateral surface of the insulating film defining the second groove.

19. A light emitting element, comprising: a semiconductor structure body comprising: a first semiconductor layer, a second semiconductor layer, an active layer positioned between the first semiconductor layer and the second semiconductor layer, a first surface, the first surface being a surface of the first semiconductor layer positioned at a side opposite to the active layer, a second surface positioned at a side opposite to the first surface, a lateral surface connecting the first surface and the second surface, the second surface including: a first region in which a portion of the first semiconductor layer is exposed from under the second semiconductor layer and the active layer, and a second region being a surface of the second semiconductor layer positioned at a side opposite to the active layer; a first electrode electrically connected with the first semiconductor layer at the first region; and a second electrode electrically connected with the second semiconductor layer at the second region, wherein: the first surface and the lateral surface are roughened surfaces, and an area of the second surface is greater than an area of the first surface.

20. The light emitting element according to claim 19, wherein: a surface roughness of the first surface is greater than a surface roughness of the lateral surface.
